EASY · PROBLEM-SOLVING AND DATA ANALYSIS · ANSWER KEY

Ratios, rates, proportional relationships, and units

03

10 answers · Rationale-rich · Teach from doc

Q1

A

A. $13t$

Choice A is correct. It's given that the ratio x to y is equivalent to the ratio 12 to t. This can be represented by $\frac{x}{y} = \frac{12}{t}$. Substituting 156 for x in this equation yields $\frac{156}{y} = \frac{12}{t}$. This can be rewritten as $12y = 156t$. Dividing both sides of this equation by 12 yields $y = 13t$. Therefore, when $x = 156$, the value of y in terms of t is $13t$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2

A

A. 9

Choice A is correct. If the object travels 108 centimeters at a speed of 12 centimeters per second, the time of travel can be determined by dividing the total distance by the speed. This results in $\frac{108 \text{ centimeters}}{12 \text{ centimeters / second}}$, which is 9 seconds.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3

A

A. $8u$

Choice A is correct. It's given that $t = 4u$. Multiplying both sides of this equation by 2 yields $2t = 2(4u)$, or $2t = 8u$.

Choice B is incorrect and may result from dividing, instead of multiplying, the right-hand side of the equation by 2. Choices C and D are incorrect and may result from calculation errors.

Q4

B

B. 2.9

Choice B is correct. It's given that the fish swam 5,104 yards and that 1 mile is equal to 1,760 yards. Therefore, the fish swam 5,104 yards$\frac{1 \text{ mile}}{1,760 \text{ yards}}$, which is equivalent to $\frac{5,104}{1,760}$ miles, or 2.9 miles.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5

9

The correct answer is 9. It's given that the customer spent $\$\,27$ to purchase oranges at $\$\,3$ per pound. Therefore, the number of pounds of oranges the customer purchased is $\$\,27\frac{1 \text{ pound}}{\$\,3}$, or 9 pounds.

Q6

A

A. 234

Choice A is correct. It's given that a special camera is used for underwater ocean research, and this camera is at a depth of 39 fathoms. It's also given that 1 fathom is equal to 6 feet. Thus, 39 fathoms is equivalent to $39 \text{ fathoms}\frac{6 \text{ feet}}{1 \text{ fathom}}$, or 234 feet. Therefore, the camera's depth, in feet, is 234.

Choice B is incorrect. This is the camera's depth, in feet, if the camera is at a depth of 19.5 fathoms.

Choice C is incorrect. This is the camera's depth, in feet, if the camera is at a depth of 7.5 fathoms.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7

14.1

The correct answer is 14.1. It's given that a participant completes the bicycle race with an average speed of 24,816 yards per hour and $1 \text{ mile} = 1{,}760 \text{ yards}$. It follows that this average speed is equivalent to $\frac{24{,}816 \text{ yards}}{1 \text{ hour}}\frac{1 \text{ mile}}{1{,}760 \text{ yards}}$, which yields $\frac{14.1 \text{ miles}}{1 \text{ hour}}$, or 14.1 miles per hour.

Q8

B

B. 675

Choice B is correct. It's given that the event will be in a 5,400-square-foot room and that there should be at least 8 square feet per person. The maximum number of people that could attend the event can be found by dividing the total square feet in the room by the minimum number of square feet needed per person, which gives $\frac{5{,}400}{8} = 675$.

Choices A and C are incorrect and may result from conceptual or computational errors. Choice D is incorrect and may result from multiplying, rather than dividing, 5,400 by 8.

Q9 B

B. 17

Choice B is correct. It's given that $1 \text{ yard} = 36 \text{ inches}$. Therefore, 612 inches is equivalent to $612 \text{ inches}\frac{1 \text{ yard}}{36 \text{ inches}}$, which can be rewritten as $\frac{612 \text{ yards}}{36}$, or 17 yards.

Choice A is incorrect. This is the number of yards that are equivalent to 2.124 inches.

Choice C is incorrect. This is the number of yards that are equivalent to 20,736 inches.

Choice D is incorrect. This is the number of yards that are equivalent to $793{,}152$ inches.

Q10 A

A. 28,000

Choice A is correct. It's given that a kangaroo has a mass of 28 kilograms and that 1 kilogram is equal to 1,000 grams. Therefore, the kangaroo's mass, in grams, is $28 \text{ kilograms}\frac{1{,}000 \text{ grams}}{1 \text{ kilogram}}$, which is equivalent to $28{,}000$ grams.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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